QuickPhone Repair Tracking System: Physical Data Flow Diagrams

MIS 4173 Sprint 3 · Team 4 · Method: Dennis, Wixom, Roth & Haley, *Systems Analysis and Design*, 8th ed., Ch. 4 and Ch. 9

Contents

1. Context diagram: the whole system as process 0 and its five external entities.
2. Level 0 diagram, complete: nine processes, seven data stores, five entities, each drawn once.
3. Level 1 diagrams for 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0 and 9.0. Process 8.0 Report Repair Status is a read-only primitive with no Level 1.

The slide deck shows the same Level 0 in four slide-sized parts; they contain no flows that are not on the complete diagram.

How the logical DFDs were made physical (Fig. 9-1)

1. **Implementation references.** Each process names how it runs (bottom band, e.g. "Django · bench PC"). Each data store is a group of tables in the PostgreSQL database, named "PostgreSQL: ..." with its tables listed. Each data flow names its medium.
2. **Human–machine boundary.** As in Figs. 9-2 and 9-4, a dashed red line around each person who uses the system's screens: Front Desk, Technician, Manager and Owner. The two manual steps in 6.0 (check photo ID, take payment on the unlinked card machine) are human action and sit inside the Front Desk's line. The Customer has no line: like the Customer Service Department in Fig. 9-2, the customer only receives a printout and an email and never uses the system's screens.
3. **System-related components.** 9.5 Import Inventory File (one-time CSV load) with its import file and error report; audit tables job_status_history and inventory_txn inside D3 and D4.
4. **Data elements.** Each flow carries the elements listed for its logical counterpart in the data flow dictionary (models document, Appendix B), under the physical field names of the ERD, plus system fields such as time stamps and who made the change.

How to read a flow name

Following Figs. 9-2 and 9-4, every line is one data flow with one noun name that says what it is and how it travels:

Name pattern	Meaning
`… Input Form`	A screen a person fills in; data goes into the system.
`… Output Form`	A screen the system shows a person.
`… Printout`, `… Email`, `… File`, `… Report`	Printed paper, the email sent from the store's account, an uploaded file, a generated report.
`PostgreSQL Record: …`	Rows read from, or a new row added to, the store the line touches.
`PostgreSQL Record Update: …`	Existing rows changed in the store the line touches.

A Level 0 flow that groups several screens (for example *Check-In Input Form*) splits into named parts on its Level 1; the table that follows lists every split, so each Level 1 balances with Level 0. Lines run only horizontally and vertically and cross at right angles (Ch. 4).

Assumptions and open items

The implementation is a Django (Python) web application with one PostgreSQL database for both stores, chosen provisionally in the Sprint 3 alternative matrix (415 of 500, against 350 for a repair-shop subscription); the choice reverses if nobody will support the system after December 9. Whether the pickup notification email is sent by staff or automatically (FR12) is open; 5.0 is drawn staff-sent.

Balancing: Level 0 flows and their Level 1 parts

Every flow into or out of a Level 0 process appears on its Level 1 diagram, under the same name or split into the named parts below (SAD 8e Ch. 4, balancing).

Level 0 flow	Level 1 flows
1.0 Schedule Appointment	
Appointment Request Input Form	Date Picker Input Form; Booking Input Form; Appointment Change Input Form
Appointment Schedule Output Form	Available Times Output Form; Booking Confirmation Output Form; Change Confirmation Output Form
DB Record: Store Hours and Technicians	Same name on Level 1
DB Record: Existing Customer	Same name on Level 1
DB Record: New Customer	Same name on Level 1
DB Record Update: Appointment	DB Record: New Appointment; DB Record Update: Appointment Change
DB Record: Booked Appointments	Same name on Level 1; also DB Record: Appointment to Change
2.0 Register Repair Job	
Check-In Input Form	Phone Search Input Form; Customer Details Input Form; Device Intake Input Form; Technician Assignment Input Form
Customer Match Output Form	Same name on Level 1
Device Tag Printout	Same name on Level 1
Claim Ticket Printout	Same name on Level 1 (to the Customer)
DB Record: Matching Customers	Same name on Level 1
DB Record Update: Customer and Device	DB Record Update: Customer; DB Record: New Device
DB Record: Today's Appointment	Same name on Level 1
DB Record Update: Appointment Checked In	Same name on Level 1
DB Record: Technician Roster	Same name on Level 1
DB Record: Store Contact	Same name on Level 1
DB Record: New Repair Job	Same name on Level 1
3.0 Record Diagnosis	
Diagnosis Input Form	Diagnosis Job Selection Input Form; Diagnosis Findings Input Form; Labor Amount Input Form

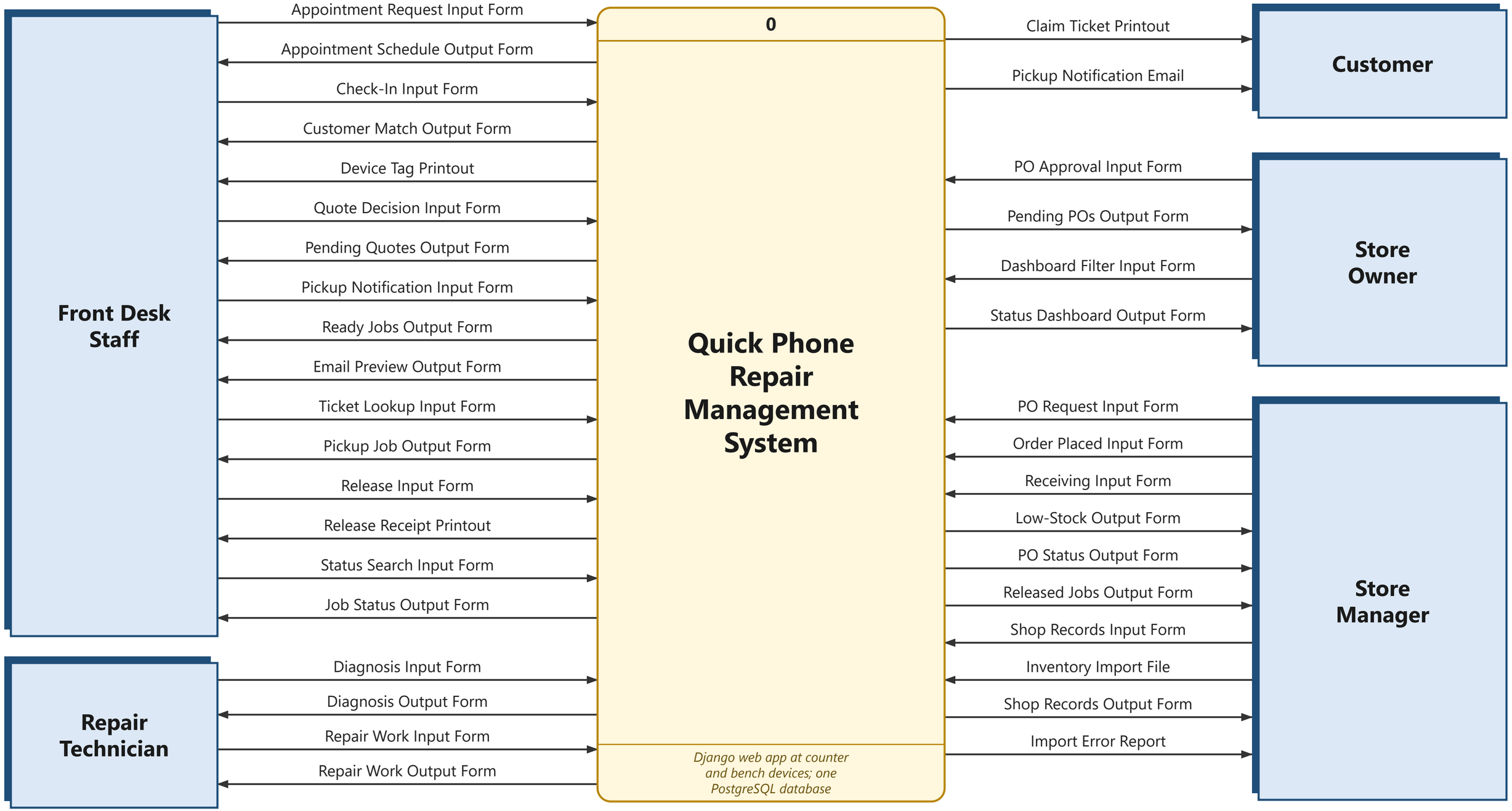
Diagnosis Output Form	Assigned Jobs Output Form; Part Availability Output Form
Quote Decision Input Form	Same name on Level 1
Pending Quotes Output Form	Same name on Level 1
DB Record: Diagnosis Queue	DB Record: Assigned Jobs; DB Record: Pending Quote
DB Record Update: Diagnosis and Quote	DB Record Update: Diagnosis Findings; DB Record: Part Needed; DB Record: New Quote; DB Record Update: Quote Decision
DB Record: Part Availability	Same name on Level 1
4.0 Record Repair Work	
Repair Work Input Form	Repair Job Selection Input Form; Parts Used Input Form; Repair Completion Input Form
Repair Work Output Form	Repair Job Output Form; Parts Shortage Output Form
DB Record: Job in Repair	Same name on Level 1
DB Record Update: Repair Work	DB Record: Parts Used; DB Record: Parts Shortage; DB Record Update: Repair Completion
DB Record: Stock on Hand	Same name on Level 1
DB Record Update: Stock Deduction	Same name on Level 1
5.0 Send Pickup Notification	
Pickup Notification Input Form	Ready Job Selection Input Form; Final Cost Input Form; Send Approval Input Form (open question FR12: removed if the email is automatic)
Ready Jobs Output Form	Same name on Level 1
Email Preview Output Form	Same name on Level 1
Pickup Notification Email	Same name on Level 1
DB Record: Ready Jobs	DB Record: Unnotified Ready Jobs; DB Record: Approved Quote Total
DB Record Update: Notice Sent	DB Record Update: Final Cost; DB Record Update: Email Sent
DB Record: Customer Email Address	Same name on Level 1
DB Record: Store Pickup Details	Same name on Level 1
6.0 Record Device Release	
Ticket Lookup Input Form	Same name on Level 1
Release Input Form	Photo ID and Claim Ticket (to manual 6.5, which gives ID Check Input Form); Payment Input Form (from manual 6.6); Signature Input Form

Pickup Job Output Form	Same name on Level 1; also Pickup Identity Output Form and Amount Due Output Form (read by staff doing 6.5 and 6.6)
Release Receipt Printout	Same name on Level 1
DB Record: Customer Name	Same name on Level 1
DB Record: Job Awaiting Pickup	Same name on Level 1
DB Record Update: Payment and Release	DB Record: New Payment; DB Record Update: Job Released
7.0 Process Purchase Orders	
PO Request Input Form	Same name on Level 1
Order Placed Input Form	Same name on Level 1
Receiving Input Form	Same name on Level 1
Low-Stock Output Form	Same name on Level 1
PO Status Output Form	Same name on Level 1
Released Jobs Output Form	Same name on Level 1
PO Approval Input Form	Same name on Level 1
Pending POs Output Form	Same name on Level 1
DB Record: Jobs Awaiting Parts	Same name on Level 1; also DB Record: Jobs Waiting on PO
DB Record Update: Awaiting-Parts Jobs	DB Record Update: Job Part Order; DB Record Update: Jobs Released
DB Record: Low-Stock Parts	Same name on Level 1
DB Record Update: Stock Received	Same name on Level 1
DB Record: Open Purchase Orders	DB Record: POs Pending Approval; DB Record: Approved POs; DB Record: Ordered PO Lines
DB Record Update: Purchase Order	DB Record: New Purchase Order; DB Record Update: PO Decision; DB Record Update: PO Placed; DB Record Update: PO Received
DB Record: Supplier for Order	Same name on Level 1
9.0 Maintain Shop Records	
Shop Records Input Form	Staff Account Input Form; Store Details Input Form; Part Catalog Input Form; Supplier Input Form
Shop Records Output Form	Staff Account Output Form; Store Details Output Form; Part Catalog Output Form; Supplier Output Form
Inventory Import File	Same name on Level 1
Import Error Report	Same name on Level 1

DB Record: Staff and Stores	DB Record: Staff Accounts; DB Record: Store Details
DB Record Update: Staff and Stores	DB Record Update: Staff Account; DB Record Update: Store Details
DB Record: Parts Catalog	Same name on Level 1; also DB Record: Existing Part Names
DB Record Update: Parts Catalog	Same name on Level 1; also DB Record: Imported Stock
DB Record: Supplier List	Same name on Level 1; also DB Record: Supplier Names
DB Record Update: Supplier List	Same name on Level 1

QuickPhone — Context Diagram (physical)

The whole system as one process and the five external entities it serves; flows match the Level 0 diagram



QuickPhone — Level 0 DFD (complete) (physical)

All nine processes, seven data stores and five external entities, each drawn once. Each line is one data flow; its name sits beside the process it leaves or enters.

Human–machine boundary (around each person who uses the system's screens)

Front Desk Staff

Quote Decision Input Form
Pending Quotes Output Form

1.0 Schedule Appointment
Django booking screen

Appointment Request Input Form
Appointment Schedule Output Form
PostgreSQL Record: Booked Appointments
PostgreSQL Record Update: Appointment
PostgreSQL Record: Existing Customer
PostgreSQL Record: New Customer
PostgreSQL Record: Store Hours and Technicians

8.0 Report Repair Status
Django dashboard

Status Search Input Form
Job Status Output Form
PostgreSQL Record: Customer Contact
PostgreSQL Record: Store List
PostgreSQL Record: Job Status History
PostgreSQL Record: Stock Summary
Dashboard Filter Input Form
Status Dashboard Output Form

6.0 Record Device Release
Django; card machine not linked

Ticket Lookup Input Form
Pickup Job Output Form
Release Input Form
Release Receipt Printout
PostgreSQL Record: Customer Name
PostgreSQL Record: Job Awaiting Pickup
PostgreSQL Record Update: Payment and Release

5.0 Send Pickup Notification
Django → store email (SMTP)

Pickup Notification Input Form
Ready Jobs Output Form
Email Preview Output Form
PostgreSQL Record: Customer Email Address
PostgreSQL Record: Store Pickup Details
PostgreSQL Record: Ready Jobs
PostgreSQL Record Update: Notice Sent

2.0 Register Repair Job
Django + label printer

Check-In Input Form
Customer Match Output Form
Device Tag Printout
PostgreSQL Record: Today's Appointment
PostgreSQL Record Update: Appointment Checked In
PostgreSQL Record: Matching Customers
PostgreSQL Record Update: Customer and Device
PostgreSQL Record: Technician Roster
PostgreSQL Record: Store Contact
PostgreSQL Record: New Repair Job

Claim Ticket Printout

Customer

Pickup Notification Email

D2 PostgreSQL: Appointments
appointment

D1 PostgreSQL: Customers & Devices
customer, device

D7 PostgreSQL: Staff & Stores
store, employee

D3 PostgreSQL: Repair Jobs
repair_job, job_status_history, quote, payment, job_part

D4 PostgreSQL: Parts Inventory
part, store_part_stock, inventory_txn

D6 PostgreSQL: Suppliers
supplier

D5 PostgreSQL: Purchase Orders
purchase_order, po_line

3.0 Record Diagnosis
Django · bench PC

PostgreSQL Record: Diagnosis Queue
PostgreSQL Record Update: Diagnosis and Quote
PostgreSQL Record: Part Availability
Diagnosis Input Form
Diagnosis Output Form

4.0 Record Repair Work
Django · bench PC

PostgreSQL Record: Job in Repair
PostgreSQL Record Update: Repair Work
PostgreSQL Record: Stock on Hand
PostgreSQL Record Update: Stock Deduction
Repair Work Input Form
Repair Work Output Form

Repair Technician

9.0 Maintain Shop Records
Django admin + CSV import

PostgreSQL Record: Staff and Stores
PostgreSQL Record Update: Staff and Stores
PostgreSQL Record: Parts Catalog
PostgreSQL Record Update: Parts Catalog
PostgreSQL Record: Supplier List
PostgreSQL Record Update: Supplier List
Shop Records Input Form
Shop Records Output Form
Inventory Import File
Import Error Report

7.0 Process Purchase Orders
Django; orders placed off-system

PostgreSQL Record: Jobs Awaiting Parts
PostgreSQL Record Update: Awaiting-Parts Jobs
PostgreSQL Record: Low-Stock Parts
PostgreSQL Record Update: Stock Received
PostgreSQL Record: Supplier for Order
PostgreSQL Record: Open Purchase Orders
PostgreSQL Record Update: Purchase Order
PO Request Input Form
Order Placed Input Form
Receiving Input Form
Low-Stock Output Form
PO Status Output Form
Released Jobs Output Form

Store Manager

PO Approval Input Form
Pending POs Output Form

Store Owner

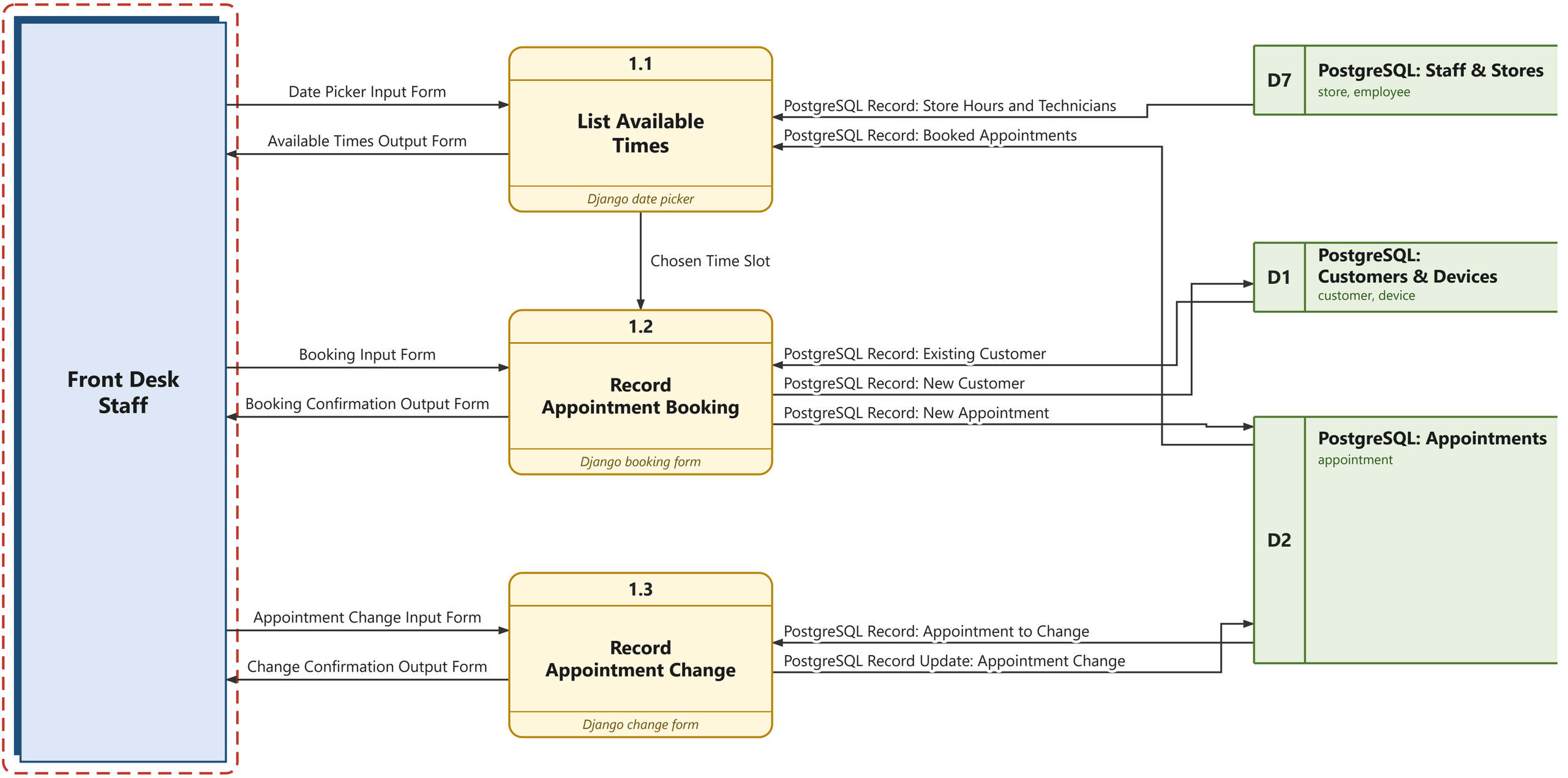
Physical DFD symbols (Gane and Sarson)

External entity
Automated process
Manual process
Data store (ERD entities listed)
Data flow (one way)
Human–machine boundary

Bottom band of a process = how it is implemented and by whom.

QuickPhone — Level 1 DFD: 1.0 Schedule Appointment (physical)

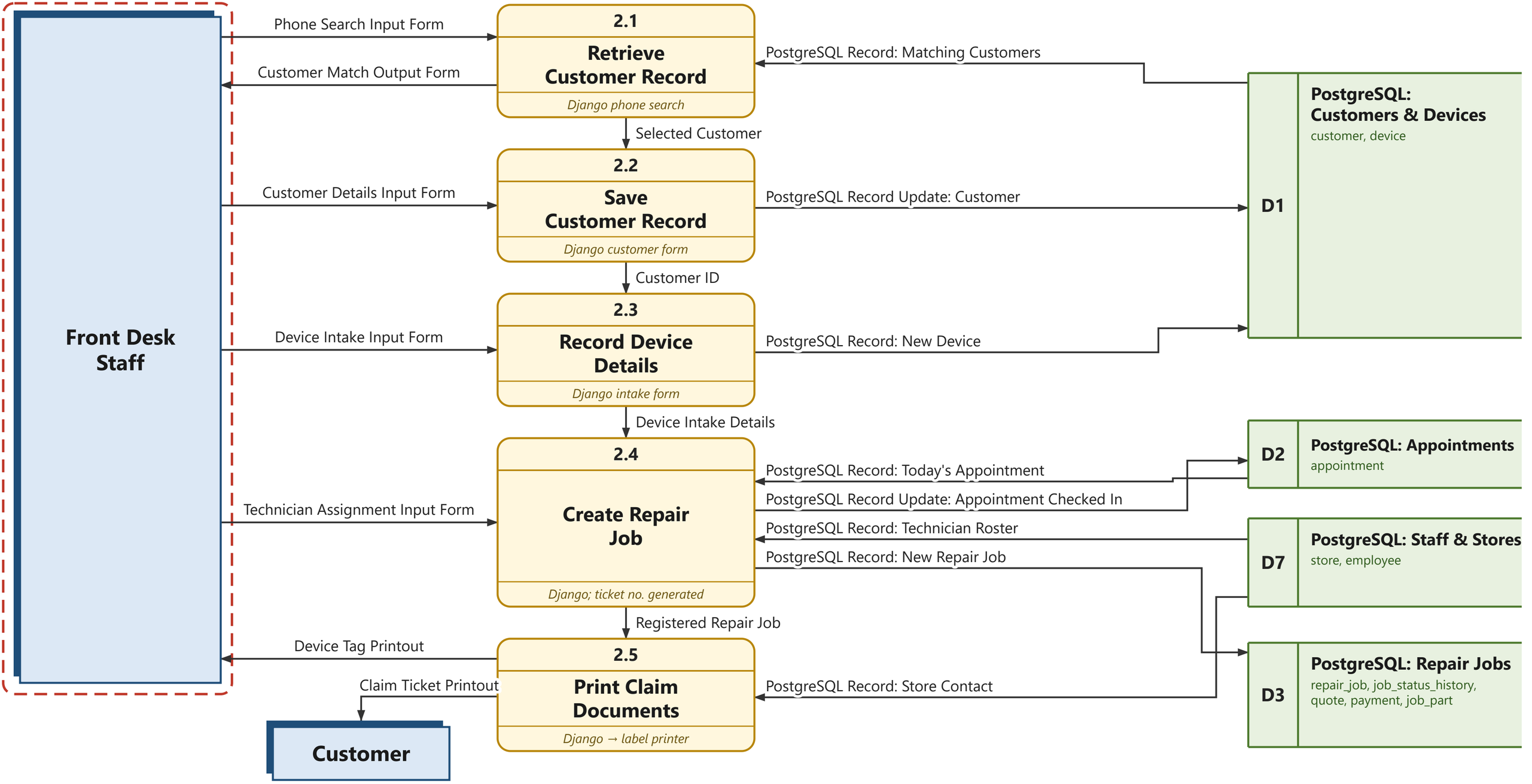
Optional appointment booking (FR01, UC-12). A walk-in needs no appointment; 2.0 pre-fills a job from one when it exists.



QuickPhone — Level 1 DFD: 2.0 Register Repair Job (physical)

Human–machine boundary (around each person who uses the system's screens)

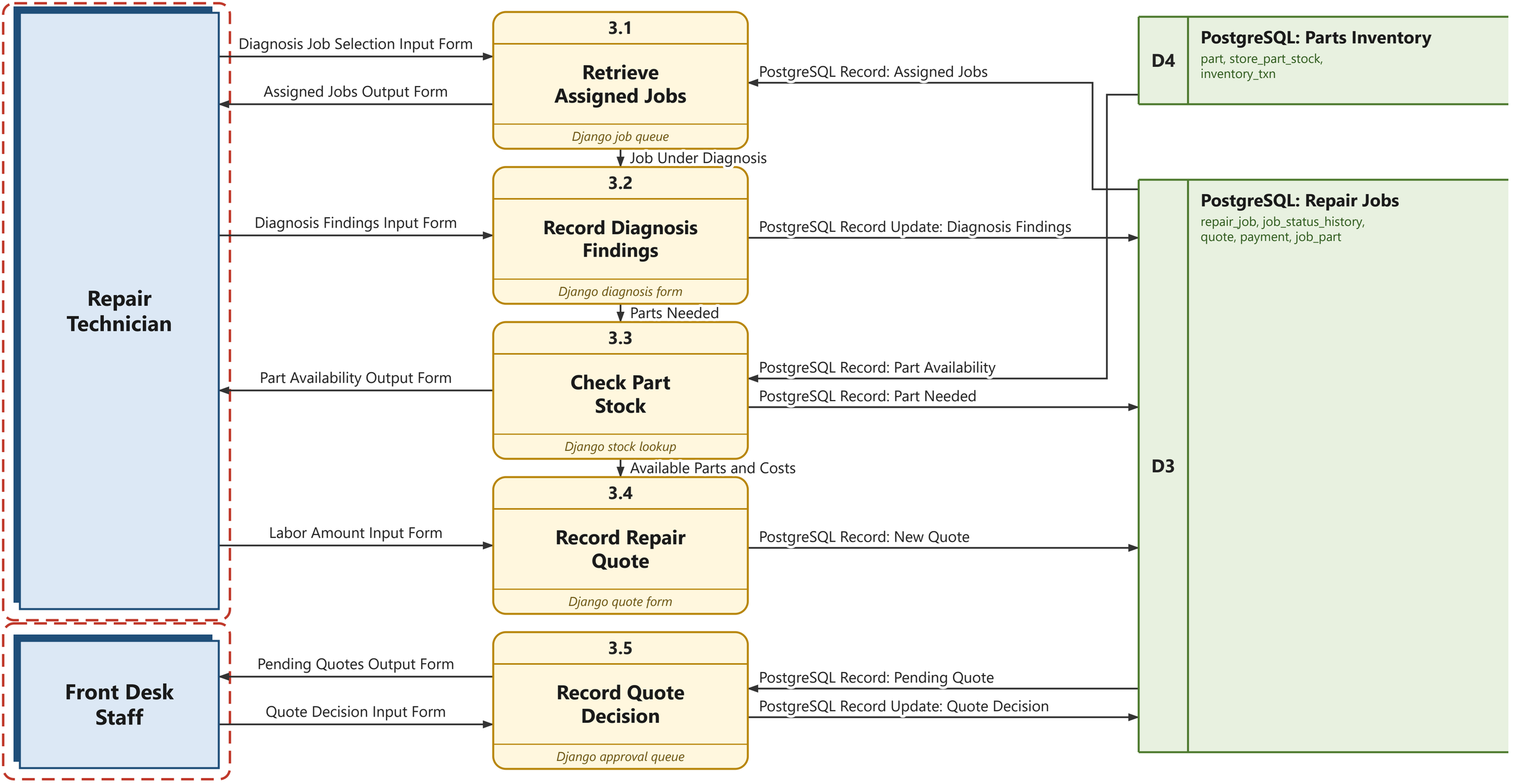
From UC-01 and FR02–FR04. 2.1 finds a returning customer by phone; 2.2 creates the record when none matches or updates the one found.



QuickPhone — Level 1 DFD: 3.0 Record Diagnosis (physical)

From UC-02 and UC-03 (FR05–FR07). The system records the diagnosis and quote; the technician does the diagnosing.

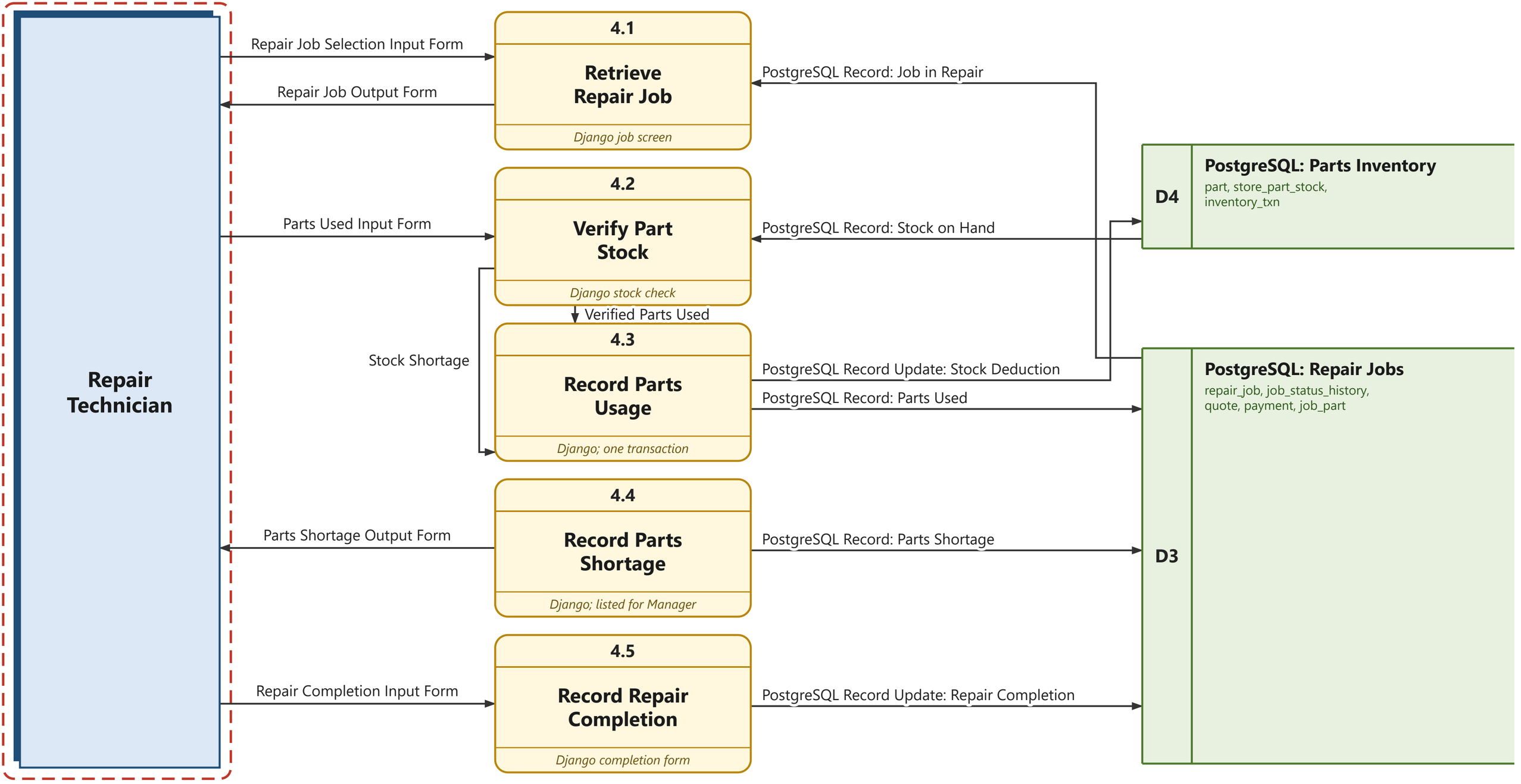
Human–machine boundary (around each person who uses the system's screens)



QuickPhone — Level 1 DFD: 4.0 Record Repair Work (physical)

From UC-04 (FR06, FR07). Recording a part as used deducts it from the store's stock automatically; there is no reservation.

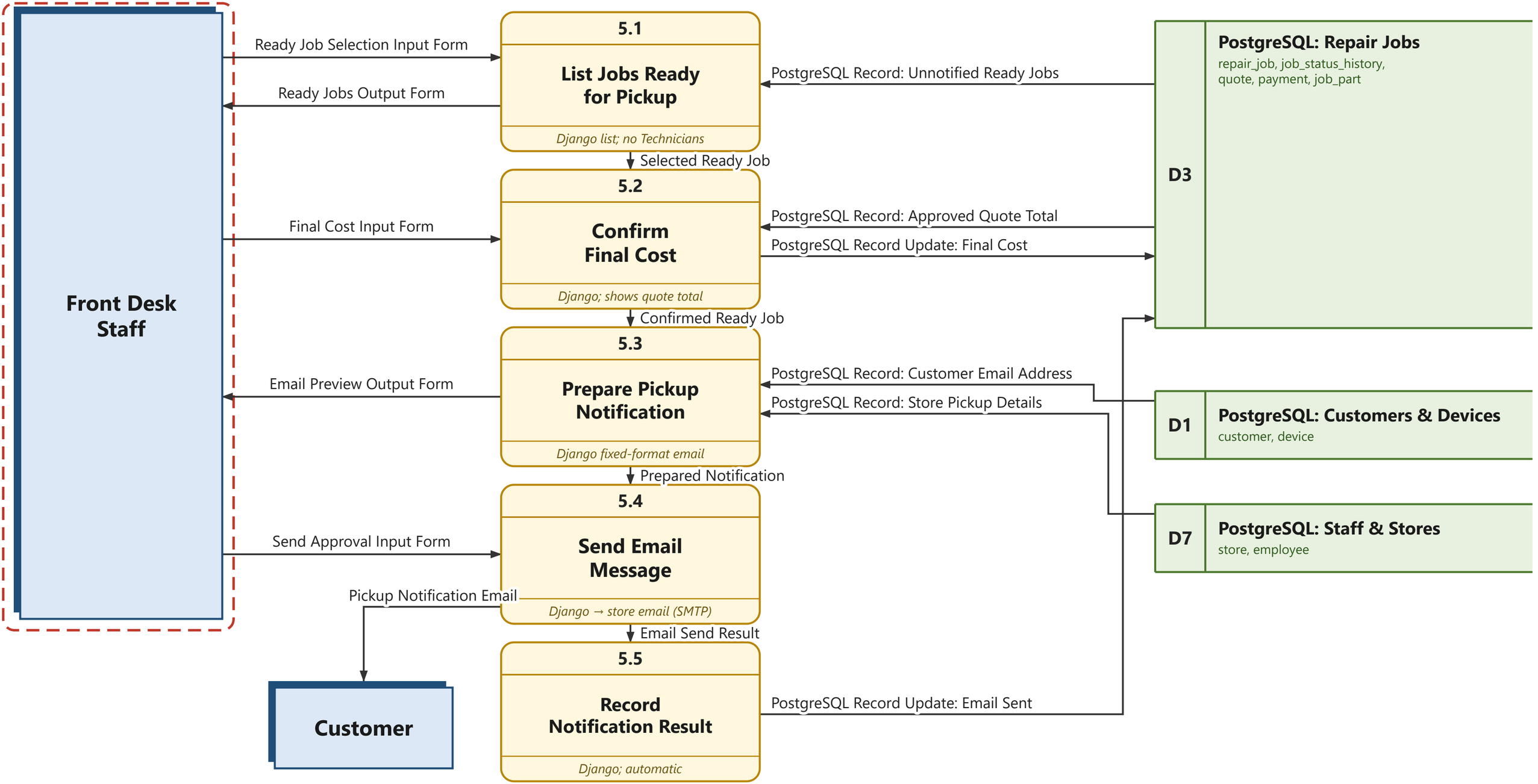
Human–machine boundary (around each person who uses the system's screens)



QuickPhone — Level 1 DFD: 5.0 Send Pickup Notification (physical)

Human–machine boundary (around each person who uses the system's screens)

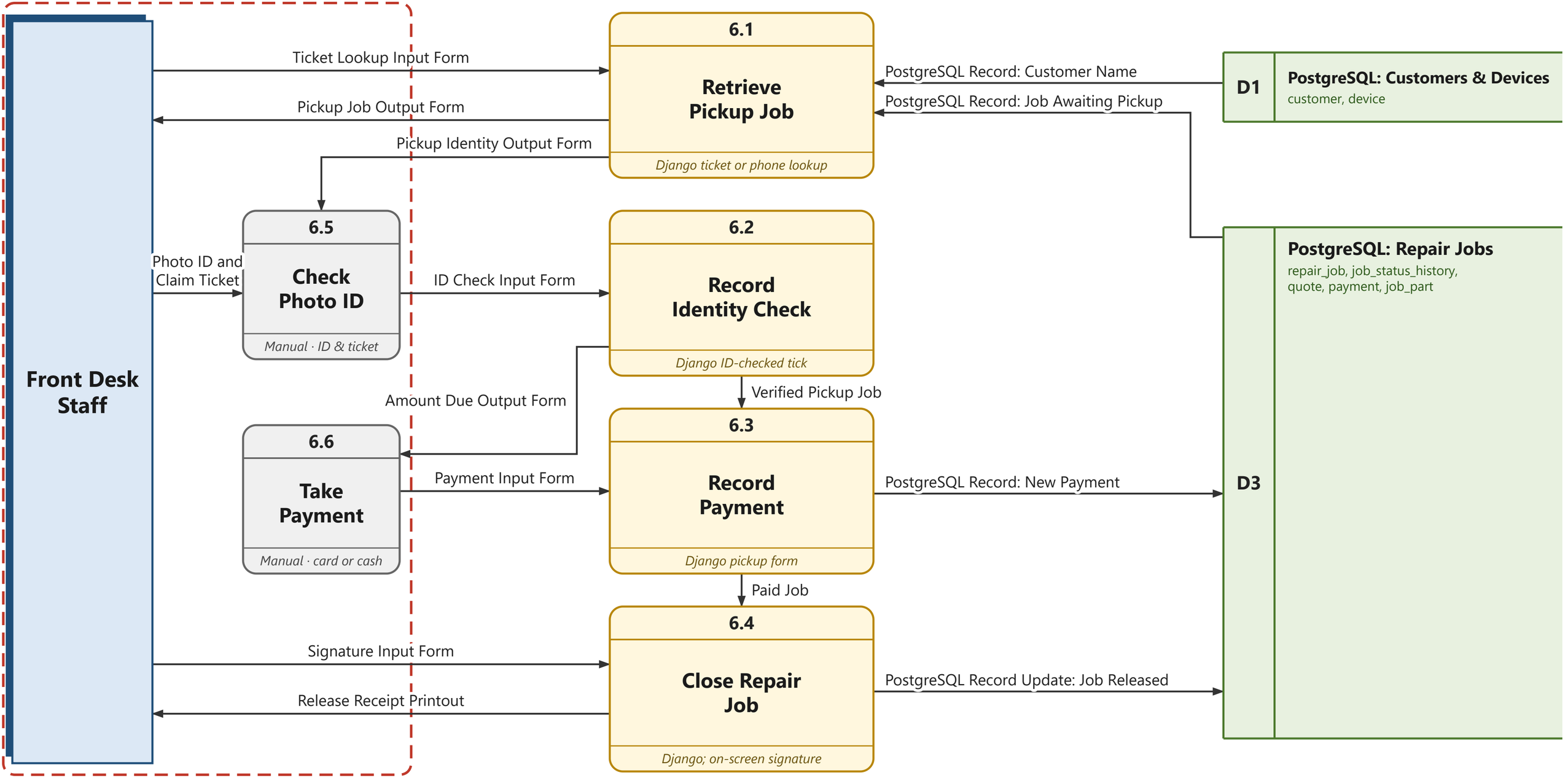
UC-05 (FR12, NFR05). Open question: staff-sent or automatic email. Drawn staff-sent; if automatic, the Send Approval Input Form is removed.



QuickPhone — Level 1 DFD: 6.0 Record Device Release (physical)

Human–machine boundary (around each person who uses the system's screens)

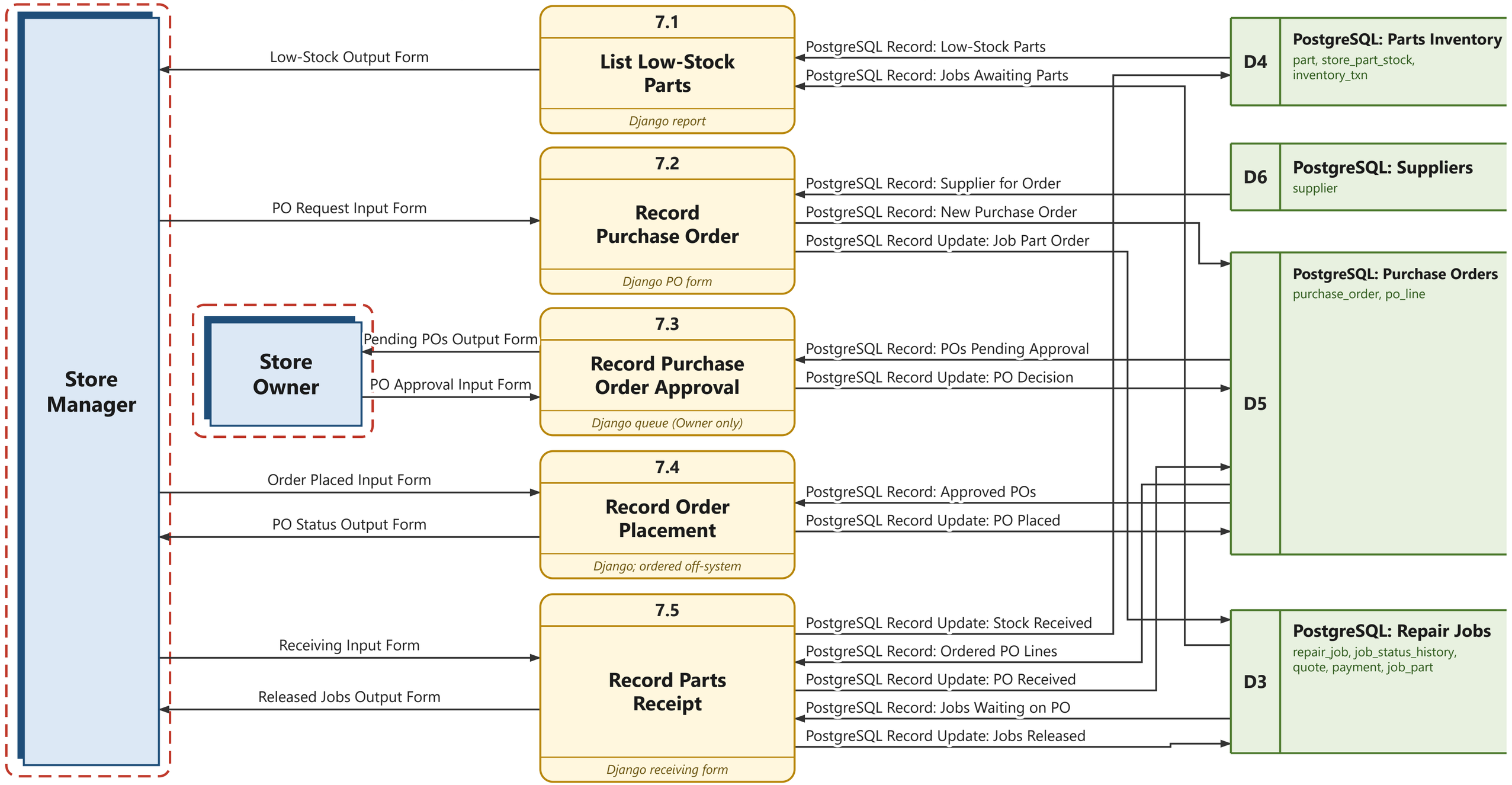
From UC-06 (FR08). The photo-ID check (6.5) and taking payment (6.6) are manual; the card machine is not linked to the system.



QuickPhone — Level 1 DFD: 7.0 Process Purchase Orders (physical)

Human–machine boundary (around each person who uses the system's screens)

From UC-08 to UC-10 (FR09, FR10). The Owner approves every purchase order; orders are placed with the supplier outside the system.



QuickPhone — Level 1 DFD: 9.0 Maintain Shop Records (physical)

From UC-07 and UC-11 (FR09, FR11, FR14). 9.5 adds the one-time CSV import of starting inventory as a batch process.

